

Nicholas M. Rapidis

Curriculum Vitae

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Education

- 2019-present **Stanford University**
Ph.D. Candidate in Physics. Expected graduation: Spring 2026
Advisor: *Prof. Kent Irwin*
M.S. in Physics completed in July 2022
- 2015-2019 **University of California, Berkeley**
B.A. in Physics. Graduated with High Distinction in General Scholarship and Departmental Honors
Thesis Title: *Resonant Axion-Photon Scattering and Galactic Searches for Axions*
Advisor: *Prof. Karl van Bibber*

Research Experience

- 2020-present **Graduate Research Assistant**, Stanford University
Advisor: *Prof. Kent Irwin*
Member of the Dark Matter Radio (DMRadio) collaboration. Lead on the first design and optimization of science reach of DMRadio-m³. Lead on DMRadio-50L SQUID readout chain, superconducting sheath design, other cryogenic hardware components.
Former lead on SQUID testing for BICEP array
- 2019-2020 **Graduate Research Assistant**, Stanford Institute for Theoretical Physics
Advisor: *Prof. Savas Dimopoulos*
Studied phenomenology of dense dark matter axion clumps (oscillons) collisions with neutron stars.
Set limits on oscillon dark matter abundance.
- 2016-2019 **Undergraduate Research Assistant**, UC Berkeley
Advisor: *Prof. Karl van Bibber*
Member of the *Haloscope at Yale Sensitive to Axion Cold Dark Matter (HAYSTAC)* collaboration.
Co-lead on refurbishment and optimization for cavity used HAYSTAC Phase II. Introduced extensive use of finite element simulation techniques for axion cavity characterization.

Honors & Awards

- 2022 *Young Scientist Award at Identification of Dark Matter 2022*: One of best three talks (out of 90) given by graduate students and postdocs at the conference.

2019 *Phi Beta Kappa*
2018-2019 *Haas Scholar*

Professional Activities

2023- Journal Referee: *Physical Review Letters*, *Physical Review D*, *Journal of Low Temperature of Physics*

Publications & Talks

Journal Articles

[[INSPIRE PROFILE](#)] [[GOOGLE SCHOLAR PROFILE](#)]

PRINCIPAL AUTHOR

- [2] **Electromagnetic modeling and science reach of DMRadio-m³**
A. AlShirawi *et al.* *Phys. Rev. D* **112**, 052001 (2025) [[arXiv:2302.14084](#)][[INSPIRE](#)]
- [1] **Characterization of the HAYSTAC axion dark matter search cavity using microwave measurement and simulation techniques**
N.M. Rapis *et al.* *Review of Scientific Instruments* **90**, 024706 (2019) [[arXiv:1809.02246](#)][[INSPIRE](#)].

JOINT PRINCIPAL AUTHOR

- [1] **Resonant Conversion of Dark Matter Oscillons in Pulsar Magnetospheres**
A. Prabhu and N.M. Rapis, *JCAP* **10**, (2020) 054 [[arXiv:2005.03700](#)][[INSPIRE](#)].

CONTRIBUTING AUTHOR

- [13] **Noise limits for dc SQUID readout of high-*Q* resonators below 300 MHz**
V. Ankel *et al.* *J. Appl. Phys.* **138**, 094505 (2025) [[arXiv:2504.20398](#)] [[INSPIRE](#)]
- [12] **Dark Matter Axion Search with HAYSTAC Phase II**
X. Bai *et al.* *Phys. Rev. Lett.* **134**, 151006 (2025) [[arXiv:2409.08998](#)] [[INSPIRE](#)]
- [11] **Measurements of DC SQUID Damping Effects on Superconducting Resonant Circuits**
E. C. van Assendelft *et al.* *IEEE Transactions on Applied Superconductivity* **33**, 5, (2023) [[INSPIRE](#)]
- [10] **New Results from HAYSTAC's Phase II Operation with a Squeezed State Receiver**
M.J. Jewell *et al.* *Phys. Rev. D* **107**, 072007, (2023) [[arXiv:2301.09721](#)][[INSPIRE](#)]
- [9] **Quantum metrology of low frequency electromagnetic modes with frequency upconverters**
S.E. Kuenstner *et al.* *Phys. Rev. Research* **7**, 013281 (2025)[[arXiv:2210.05576](#)][[INSPIRE](#)]
- [8] **DMRadio-m³: A Search for the QCD Axion Below 1 μ eV**
L. Brouwer *et al.* *Phys. Rev. D* **106**, 103008, (2022) [[arXiv:2204.13781](#)][[INSPIRE](#)]
- [7] **Introducing DMRadio-GUT, a search for GUT-scale QCD axions**
L. Brouwer *et al.* *Phys. Rev. D* **106**, 112003, (2022) [[arXiv:2203.11246](#)][[INSPIRE](#)]
- [6] **A Model-Independent Radio Telescope Dark Matter Search**
A. Keller, *et al.* *Astrophys. J.* **927** (2022) 1, 71. [[arXiv:2112.03439](#)][[INSPIRE](#)]

- [5] **A quantum-enhanced search for dark matter axions**
K.M. Backes *et al.* *Nature* **590**, 238-242 (2021) [[arXiv:2008.01853](#)][INSPIRE]
- [4] **An improved analysis framework for axion dark matter searches**
D.A. Palken *et al.* *Phys. Rev. D* **101**, 123011, (2020) [[arXiv:2003.08510](#)][INSPIRE].
- [3] **Results from Phase 1 of the HAYSTAC microwave cavity axion experiment**
L. Zhong *et al.* *Phys. Rev. D* **97**, 092001, (2018) [[arXiv:1803.03690](#)][INSPIRE].
- [2] **Design and Operational Experience of a Microwave Cavity Axion Detector for the 20-100 μeV Range**
S. Al Kenany *et al.* *Nuclear Instruments and Methods in Physics Research A* **854** (2017) 11–24. [[arXiv:1611.07123](#)] [INSPIRE].
- [1] **First Results from a Microwave Cavity Axion Search at 24 μeV**
B.M. Brubaker *et al.* *Phys. Rev. Lett.* **118**, 061302 (2017) [[arXiv:1610.02580](#)][INSPIRE].

Talks

INVITED TALKS

- [2] **Searching for low mass dark matter axions with DMRadio**
Physics Seminar, Boston University, July 24, 2024, Boston, MA
- [1] **Status of the DMRadio Program**
YOUNGST@RS - Shoot for the Stars, Aim for the Axions, October 4-7, 2022, Virtual.

CONTRIBUTED TALKS

- [9] **Overview of the DMRadio Series of Experiments**
APS Global Physics Summit, March 16-21, 2024, Anaheim, CA
- [8] **Science reach and electromagnetic modeling of DMRadio- m^3**
APS April Meeting 2024, April 3-6, 2024, Sacramento, CA
- [7] **Science reach and electromagnetic modeling of DMRadio- m^3**
Topics in Astroparticle and Underground Physics, Aug 28-Sep 1, 2023, Vienna, Austria
- [6] **Status of DMRadio 50L and m^3**
Identification of Dark Matter, July 18-22, 2022, Vienna, Austria
- [5] **Modeling and optimizing DMRadio using an equivalent circuit formalism**
APS April Meeting 2021, April 17-20, 2021, Virtual
- [4] **Electromagnetic sensing below the Standard Quantum Limit: 3 kHz to 300 MHz**
APS March Meeting 2021, March 15-19, 2021, Virtual
- [3] **Characterization of the HAYSTAC dark matter detector cavity: microwave measurement and simulation**
APS April Meeting 2019, April 13-16, 2019, Denver, CO
- [2] **Completion of Phase I and Preparation for Phase II of the HAYSTAC Experiment**
14th Patras Workshop on Axions, WIMPs, and WISPs, June 18-22, 2018, DESY, Hamburg, Germany
- [1] **Application of the Bead Perturbation Technique to a Study of a Tunable 5 GHz Annular Cavity**
2nd Workshop on Microwave Cavities and Detectors for Axion Research, January 10-13, 2017, LLNL, Livermore, CA

Teaching Experience

Head Teaching Assistant, Stanford University

Spr. 2022 Physics 25 – Modern Physics (Instructor: Kent Irwin).

2021-2022 **Mentor**, Polygence

One-on-one mentoring of high school students on research projects in their pre-collegiate schooling. Projects topics in dark matter physics and cosmology.

Teaching Assistant, Stanford University

Fall 2020 Physics 46 – Heat and Optics (Instructor: Giorgio Gratta).

Spr. 2020 Physics 43 – Electricity and Magnetism (Instructor: Mark Kasevich).

Sum. 2017 **Reader (Grader)**, UC Berkeley

Physics 137A – Quantum Mechanics I

Skills

Programming & Software

Languages: *Python*, *Mathematica*, *Matlab*, *LabVIEW*

Software: *COMSOL (AC/DC, RF, Heat Transfer)*, *Fusion 360*, *CST Microwave Studio*, *KiCad*, *FastHenry*

Other: $\text{\LaTeX}$, HTML

Experimental Tools

Dilution refrigerators, Liquid cryogen dips, Network analyzers, Lock-in amplifiers, Wirebonding, Machine shop skills, 3D printing

Languages

English (native), Greek (native), German (advanced proficiency)